# AER309 Basic Thermodynamics

Term Test 1 practice B

Chapters 1 to 5 | 80 minutes | 100 marks

Closed book, with a calculator and one letter-sized, one-sided formula sheet. Show all working; no marks are given for an answer alone. Questions 2, 5 and 6 need the water tables A-4, A-5 and A-6.

**Constants.** Air: $R = 0.287$, $c_p = 1.005$, $c_v = 0.718$ kJ/(kg·K). Liquid water: $c = 4.18$ kJ/(kg·K). Mercury: $\rho = 13\,600$ kg/m$^3$. $\sigma = 5.67 \times 10^{-8}$ W/(m$^2$·K$^4$). $g = 9.81$ m/s$^2$.

Each question is on its own page with space for your working; the worked solution starts on the next page. Answers are given to four significant figures. Sign convention: $Q > 0$ into the system, $W > 0$ out of it.

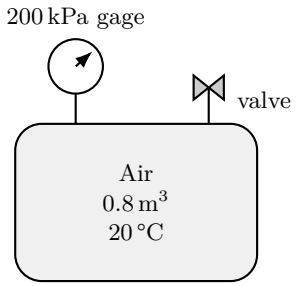
## Question 1 [10 marks]

A rigid $0.8\,\mathrm{m}^3$ tank holds air at $20\,^\circ\mathrm{C}$. The pressure gauge on the tank reads 200 kPa and the atmospheric pressure is 98 kPa.

(a) Find the mass of air in the tank.

(b) The tank is left in the sun and the air warms to $60\,^\circ\mathrm{C}$. Find the new gauge reading.

(c) A valve is opened to bring the gauge back to 200 kPa while the air stays at $60\,^\circ\mathrm{C}$. Find the mass of air that has to be let out.



## Solution to Question 1

**Given** $V = 0.8\,\mathrm{m}^3$, $T_1 = 20\,^\circ\mathrm{C} = 293.15\,\mathrm{K}$, $P_{\mathrm{gage},1} = 200\,\mathrm{kPa}$, $P_{\mathrm{atm}} = 98\,\mathrm{kPa}$, $T_2 = 60\,^\circ\mathrm{C} = 333.15\,\mathrm{K}$.

**Find** ($m_1$, $P_{\mathrm{gage},2}$, $m_{\mathrm{out}}$)

**Strategy** The ideal-gas law needs absolute pressure and absolute temperature. Convert first, then use $PV = mRT$ at each state.

**Assumptions** Air is an ideal gas. The tank is rigid, so $V$ is fixed.

**Analysis** *(a) Mass.* The absolute pressure is $P_1 = P_{\mathrm{gage},1} + P_{\mathrm{atm}} = 200 + 98 = 298\,\mathrm{kPa}$.

$$m_1 = \frac{P_1 V}{R T_1} = \frac{(298\,\mathrm{kPa})(0.8\,\mathrm{m}^3)}{(0.287\,\mathrm{kJ/(kg\,K)})(293.15\,\mathrm{K})} = 2.8336\,\mathrm{kg}.$$

*(b) Heating at constant volume and mass.* With $m$ and $V$ fixed, $P/T$ is constant:

$$P_2 = P_1 \frac{T_2}{T_1} = (298\,\mathrm{kPa})\,\frac{333.15}{293.15} = 338.66\,\mathrm{kPa},$$

$$P_{\mathrm{gage},2} = P_2 - P_{\mathrm{atm}} = 338.66 - 98 = 240.66\,\mathrm{kPa}.$$

*(c) Bleeding air.* State 3 is back at $P_3 = 298\,\mathrm{kPa}$ with $T_3 = 333.15\,\mathrm{K}$:

$$m_3 = \frac{P_3 V}{R T_3} = \frac{(298)(0.8)}{(0.287)(333.15)}\,\mathrm{kg} = 2.4934\,\mathrm{kg},$$

$$m_{\mathrm{out}} = m_1 - m_3 = 2.8336 - 2.4934 = 0.3402\,\mathrm{kg}.$$

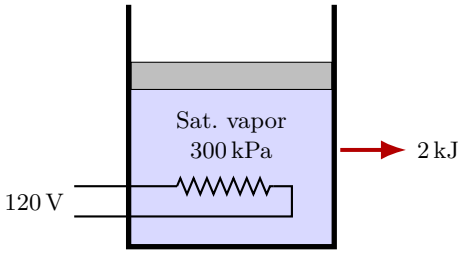
| | | |
|---|---|---|
| (a) $m_1 = 2.834\,\mathrm{kg}$ | (b) the gauge reads $240.7\,\mathrm{kPa}$ | (c) $m_{\mathrm{out}} = 0.3402\,\mathrm{kg}$ |

**Discussion** Using gage pressure and °C in part (b) gives $200 \times 60/20 = 600\,\mathrm{kPa}$, which is wildly off. The absolute temperature rises by only 13.6 %, so the absolute pressure rises by 13.6 % too.

## Question 2 [20 marks]

A piston-cylinder device contains 0.05 kg of saturated water vapor at 300 kPa. The piston is free to move, so the pressure stays constant. A resistance heater inside the cylinder is connected to a 120 V supply and draws 0.5 A for 4 min. During that time the steam loses 2 kJ of heat through the cylinder wall.

(a) Find the electrical work done on the steam.
(b) Find the final temperature of the steam.
(c) Find the boundary work.



## Solution to Question 2

**Sign convention** $Q > 0$ into the steam, $W > 0$ done by the steam. The heater does work on the steam, so it enters the balance as $-W_{\text{e,in}}$.

**Given** $m = 0.05\,\text{kg}$, $P = 300\,\text{kPa}$ (constant), state 1 saturated vapor, $\mathcal{E} = 120\,\text{V}$, $I = 0.5\,\text{A}$, $\Delta t = 4\,\text{min} = 240\,\text{s}$, $Q_{\text{out}} = 2\,\text{kJ}$.

**Find** $(W_{\text{e,in}},\, T_2,\, W_{\text{b}})$

**Strategy** At constant pressure the boundary work and $\Delta U$ combine into $\Delta H$, so the energy balance gives $h_2$ directly. Then $P$ and $h_2$ fix state 2 in Table A-6.

**Assumptions** Closed system: the steam. The heater wire itself is part of the system and stores no energy. Quasi-equilibrium expansion at constant pressure. Kinetic and potential energy changes are zero.

**Analysis** *(a) Electrical work.*

$$W_{\text{e,in}} = \mathcal{E} I\, \Delta t = (120\,\text{V})(0.5\,\text{A})(240\,\text{s}) = 14\,400\,\text{J} = 14.40\,\text{kJ}.$$

*(b) Energy balance.*

$$Q - W = \Delta U \quad \Longrightarrow \quad -Q_{\text{out}} - (W_{\text{b}} - W_{\text{e,in}}) = U_2 - U_1.$$

At constant pressure $W_{\text{b}} = P(V_2 - V_1)$, and $\Delta U + P\,\Delta V = \Delta H$, so

$$W_{\text{e,in}} - Q_{\text{out}} = m\,(h_2 - h_1) \quad \Longrightarrow \quad h_2 = h_1 + \frac{W_{\text{e,in}} - Q_{\text{out}}}{m}.$$

State 1 is saturated vapor at 300 kPa. From Table A-5, $h_1 = h_{\text{g}} = 2\,724.9\,\text{kJ/kg}$ and $v_1 = v_{\text{g}} = 0.60582\,\text{m}^3\text{/kg}$. Then

$$h_2 = 2724.9 + \frac{(14.40 - 2.00)\,\text{kJ}}{0.05\,\text{kg}} = 2724.9 + 248.0 = 2\,972.9\,\text{kJ/kg}.$$

Since $h_2 > h_{\text{g}}$, state 2 is superheated vapor. Table A-6 at 0.30 MPa brackets it:

| $T$ (°C) | $v$ ($\text{m}^3$/kg) | $h$ (kJ/kg) |
|---|---|---|
| 250 | 0.79645 | 2 967.9 |
| 300 | 0.87535 | 3 069.6 |

Interpolating on $h$, the fraction of the way from the first row to the second is

$$f = \frac{2972.9 - 2967.9}{3069.6 - 2967.9} = \frac{5.0}{101.7} = 0.04916,$$

$$T_2 = 250 + (0.04916)(300 - 250) = 252.46\,^{\circ}\text{C}.$$

*(c) Boundary work.* The same fraction gives the specific volume:

$$v_2 = 0.79645 + (0.04916)(0.87535 - 0.79645) = 0.80033\,\text{m}^3\text{/kg},$$

$$W_\mathrm{b} = mP\,(v_2 - v_1) = (0.05\,\mathrm{kg})(300\,\mathrm{kPa})(0.80033 - 0.60582)\,\mathrm{m^3/kg} = 2.918\,\mathrm{kJ}.$$

| | | |
|---|---|---|
| (a) $W_{\mathrm{e,in}} = 14.40\,\mathrm{kJ}$, done on the steam | (b) $T_2 = 252.5\,^\circ\mathrm{C}$ | (c) $W_\mathrm{b} = 2.918\,\mathrm{kJ}$, done by the steam |

**Discussion** Of the net 12.40 kJ added, 2.918 kJ goes into pushing the piston up and the other 9.482 kJ raises the internal energy. Using $\Delta U$ in place of $\Delta H$ in the balance would count the boundary work twice or not at all, and it is the most common slip in this kind of question.

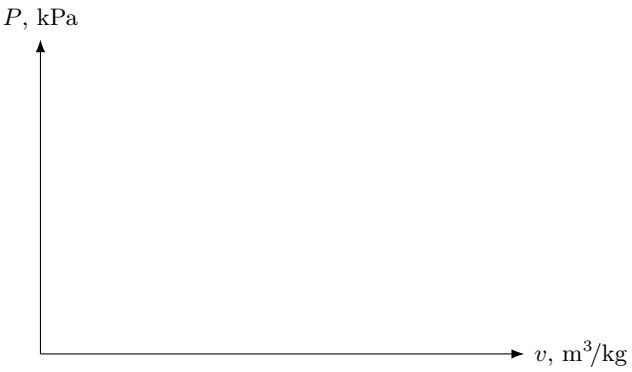
## Question 3 [15 marks]

Air in a piston-cylinder device goes through a cycle made of three processes.

- 1-2: heat is added at constant volume, from 100 kPa and 300 K until the pressure is 300 kPa.
- 2-3: the air expands with no heat transfer, following $Pv^k$ = constant with $k = 1.4$, until the pressure is back to 100 kPa.
- 3-1: the air is cooled at constant pressure back to the initial state.

Use constant specific heats.

(a) Sketch the cycle on a $P$-$v$ diagram.
(b) Find the heat added and the heat rejected, per kilogram of air.
(c) Find the net work per kilogram and the thermal efficiency.



## Solution to Question 3

**Sign convention** $Q > 0$ into the air, $W > 0$ done by the air. The magnitudes $q_{\text{in}}$ and $q_{\text{out}}$ are both written as positive numbers.

**Given** $P_1 = 100\,\text{kPa}$, $T_1 = 300\,\text{K}$, $P_2 = 300\,\text{kPa}$, $v_2 = v_1$, $P_3 = P_1$, $k = 1.4$.

**Find** ($q_{\text{in}}$, $q_{\text{out}}$, $w_{\text{net}}$, $\eta_{\text{th}}$)

**Strategy** Find the three temperatures first. Then heat is added only in 1-2 and rejected only in 3-1, and over a cycle $w_{\text{net}} = q_{\text{in}} - q_{\text{out}}$.

**Assumptions** Closed system. Ideal gas with constant specific heats. Quasi-equilibrium processes.

**Analysis** *Temperatures.* Process 1-2 is at constant volume, so $P/T$ is constant:

$$T_2 = T_1 \frac{P_2}{P_1} = (300\,\text{K})\frac{300}{100} = 900\,\text{K}.$$

Process 2-3 is polytropic with $n = k$:

$$T_3 = T_2\left(\frac{P_3}{P_2}\right)^{(k-1)/k} = (900\,\text{K})\left(\frac{100}{300}\right)^{0.4/1.4} = (900\,\text{K})(0.73060) = 657.54\,\text{K}.$$

*(b) Heat added, process 1-2.* No boundary work at constant volume, so $q - 0 = \Delta u$:

$$q_{\text{in}} = c_v\,(T_2 - T_1) = (0.718\,\text{kJ/(kg K)})(900 - 300)\,\text{K} = 430.80\,\text{kJ/kg}.$$

*Heat rejected, process 3-1.* At constant pressure $q = \Delta h$, which is negative here because the air cools. Its magnitude is

$$q_{\text{out}} = c_p\,(T_3 - T_1) = (1.005\,\text{kJ/(kg K)})(657.54 - 300)\,\text{K} = 359.33\,\text{kJ/kg}.$$

*(c) Net work and efficiency.* Over the cycle $\Delta u = 0$, so

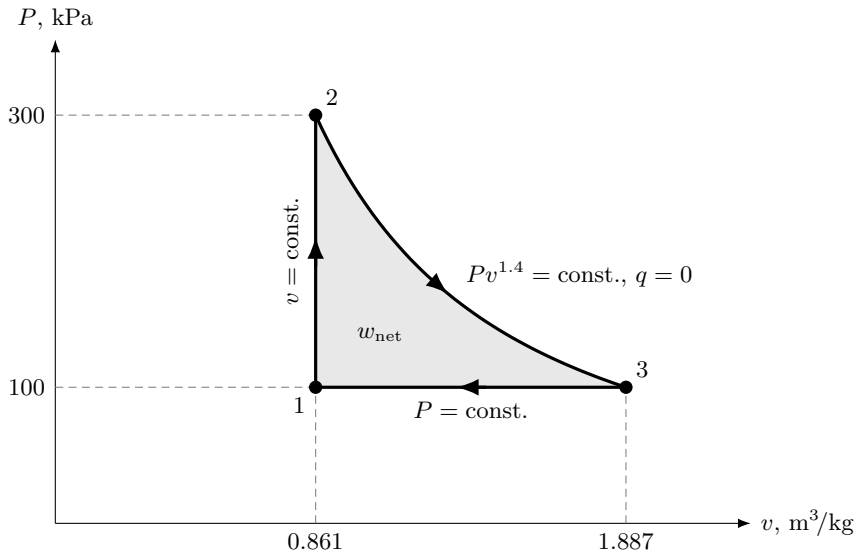
$$w_{\text{net}} = q_{\text{in}} - q_{\text{out}} = 430.80 - 359.33 = 71.47\,\text{kJ/kg},$$

$$\eta_{\text{th}} = \frac{w_{\text{net}}}{q_{\text{in}}} = \frac{71.47}{430.80} = 0.1659.$$

| |
|---|
| (b) $q_{\text{in}} = 430.8\,\text{kJ/kg}$, $q_{\text{out}} = 359.3\,\text{kJ/kg}$ (c) $w_{\text{net}} = 71.47\,\text{kJ/kg}$ done by the air, $\eta_{\text{th}} = 0.1659 = 16.59\,\%$ |

**Check** Add up the work process by process. $w_{12} = 0$. In 2-3, $q = 0$ so $w_{23} = -\Delta u = c_v(T_2 - T_3) = (0.718)(242.46) = 174.09\,\text{kJ/kg}$. In 3-1, $w_{31} = P_1(v_1 - v_3) = R(T_1 - T_3) = (0.287)(-357.54) = -102.61\,\text{kJ/kg}$. The sum is $71.47\,\text{kJ/kg}$, the same as before. The diagram is on the next page.

**$P$-$v$ diagram for Question 3** (drawn to scale)



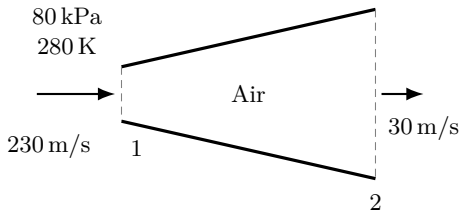

| State | $P$ (kPa) | $T$ (K) | $v$ (m$^3$/kg) |
|---|---|---|---|
| 1 | 100 | 300 | 0.861 |
| 2 | 300 | 900 | 0.861 |
| 3 | 100 | 657.54 | 1.887 |

The specific volumes come from $v = RT/P$. The cycle runs clockwise, so the net work is positive and equals the shaded area.

## Question 4 [15 marks]

Air enters the diffuser of a jet engine steadily at 80 kPa and 280 K with a velocity of 230 m/s. The inlet area is $0.04\,m^2$. The air leaves at 30 m/s through an exit area of $0.25\,m^2$. The diffuser is adiabatic.

(a) Find the mass flow rate.
(b) Find the exit temperature.
(c) Find the exit pressure.



## Solution to Question 4

**Sign convention** The balance is written as energy in equals energy out.

**Given** $P_1 = 80\,\text{kPa}$, $T_1 = 280\,\text{K}$, $V_1 = 230\,\text{m/s}$, $A_1 = 0.04\,\text{m}^2$, $V_2 = 30\,\text{m/s}$, $A_2 = 0.25\,\text{m}^2$.

**Find** $(\dot{m},\, T_2,\, P_2)$

**Assumptions** Control volume: the diffuser, one inlet and one exit. Steady flow. Adiabatic and no shaft, so $\dot{Q} = \dot{W} = 0$. Potential energy change is zero. Air is an ideal gas with constant $c_p$.

**Analysis** *(a) Mass balance at the inlet.*

$$v_1 = \frac{RT_1}{P_1} = \frac{(0.287\,\text{kJ/(kg K)})(280\,\text{K})}{80\,\text{kPa}} = 1.0045\,\text{m}^3/\text{kg},$$

$$\dot{m} = \frac{V_1 A_1}{v_1} = \frac{(230\,\text{m/s})(0.04\,\text{m}^2)}{1.0045\,\text{m}^3/\text{kg}} = 9.1588\,\text{kg/s}.$$

*(b) Energy balance.* Per unit mass,

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} \quad\Longrightarrow\quad c_p\,(T_2 - T_1) = \frac{V_1^2 - V_2^2}{2}.$$

The kinetic energy term, converted with $1\,\text{kJ/kg} = 1\,000\,\text{m}^2/\text{s}^2$:

$$\frac{V_1^2 - V_2^2}{2} = \frac{(230^2 - 30^2)\,\text{m}^2/\text{s}^2}{2}\left(\frac{1\,\text{kJ/kg}}{1\,000\,\text{m}^2/\text{s}^2}\right) = 26.00\,\text{kJ/kg}.$$

$$T_2 = T_1 + \frac{26.00\,\text{kJ/kg}}{c_p} = 280 + \frac{26.00}{1.005} = 305.87\,\text{K}.$$

*(c) Mass balance at the exit.* The same $\dot{m}$ leaves, so

$$v_2 = \frac{V_2 A_2}{\dot{m}} = \frac{(30\,\text{m/s})(0.25\,\text{m}^2)}{9.1588\,\text{kg/s}} = 0.81889\,\text{m}^3/\text{kg},$$

$$P_2 = \frac{RT_2}{v_2} = \frac{(0.287)(305.87)}{0.81889}\,\text{kPa} = 107.2\,\text{kPa}.$$

| (a) $\dot{m} = 9.159\,\text{kg/s}$ | (b) $T_2 = 305.9\,\text{K}$ | (c) $P_2 = 107.2\,\text{kPa}$ |
|---|---|---|

**Discussion** A diffuser trades velocity for pressure: the air slows down, and both its temperature and its pressure rise. Forgetting the factor of 1000 gives a temperature rise of 25 870 K in place of 25.87 K, so an absurd $T_2$ is the sign that the conversion was missed.

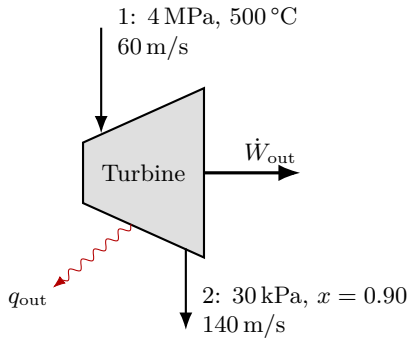
## Question 5 [20 marks]

Steam enters a turbine steadily at 4 MPa and 500 °C with a velocity of 60 m/s. It leaves at 30 kPa with a quality of 0.90 and a velocity of 140 m/s. The mass flow rate is 8 kg/s, and the turbine casing loses 25 kJ of heat for every kilogram of steam.

(a) Find the change in kinetic energy per kilogram of steam.

(b) Find the power output of the turbine.

(c) Find the inlet flow area.



## Solution to Question 5

**Sign convention** Each term is a positive quantity with its direction in the subscript, and the balance is written as energy in equals energy out.

**Given** $P_1 = 4\,\text{MPa}$, $T_1 = 500\,^\circ\text{C}$, $V_1 = 60\,\text{m/s}$, $P_2 = 30\,\text{kPa}$, $x_2 = 0.90$, $V_2 = 140\,\text{m/s}$, $\dot{m} = 8\,\text{kg/s}$, $q_{\text{out}} = 25\,\text{kJ/kg}$.

**Find** $(\Delta ke,\ \dot{W}_{\text{out}},\ A_1)$

**Assumptions** Control volume: the turbine, one inlet and one exit. Steady flow. Potential energy change is negligible.

**Properties** *State 1.* At 4 MPa the saturation temperature is 250.35 °C, so 500 °C is superheated. From Table A-6:

$$v_1 = 0.08644\,\text{m}^3/\text{kg}, \qquad h_1 = 3\,446.0\,\text{kJ/kg}.$$

*State 2.* Saturated mixture at 30 kPa. From Table A-5, $h_{\text{f}} = 289.27\,\text{kJ/kg}$ and $h_{\text{fg}} = 2\,335.3\,\text{kJ/kg}$, so

$$h_2 = h_{\text{f}} + x_2 h_{\text{fg}} = 289.27 + (0.90)(2335.3) = 2\,391.04\,\text{kJ/kg}.$$

**Analysis** *(a) Kinetic energy change.*

$$\Delta ke = \frac{V_2^2 - V_1^2}{2} = \frac{(140^2 - 60^2)\,\text{m}^2/\text{s}^2}{2}\left(\frac{1\,\text{kJ/kg}}{1\,000\,\text{m}^2/\text{s}^2}\right) = 8.000\,\text{kJ/kg}.$$

*(b) Energy balance.* Per unit mass, energy in equals energy out:

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} + w_{\text{out}} + q_{\text{out}} \quad \Longrightarrow \quad w_{\text{out}} = (h_1 - h_2) - \Delta ke - q_{\text{out}}.$$

$$w_{\text{out}} = (3446.0 - 2391.04) - 8.000 - 25 = 1054.96 - 33.00 = 1\,021.96\,\text{kJ/kg},$$

$$\dot{W}_{\text{out}} = \dot{m}\, w_{\text{out}} = (8\,\text{kg/s})(1\,021.96\,\text{kJ/kg}) = 8\,175.7\,\text{kW}.$$

*(c) Inlet area.* From $\dot{m} = V_1 A_1 / v_1$,

$$A_1 = \frac{\dot{m} v_1}{V_1} = \frac{(8\,\text{kg/s})(0.08644\,\text{m}^3/\text{kg})}{60\,\text{m/s}} = 0.011525\,\text{m}^2.$$

| |
|---|
| (a) $\Delta ke = 8.000\,\text{kJ/kg}$ (an increase) (b) $\dot{W}_{\text{out}} = 8\,176\,\text{kW} = 8.176\,\text{MW}$ (c) $A_1 = 0.01153\,\text{m}^2 = 115.3\,\text{cm}^2$ |

**Discussion** The enthalpy drop of 1 055 kJ/kg does three jobs: it speeds the steam up (8 kJ/kg), it leaks out as heat (25 kJ/kg), and what is left turns the shaft. Both losses come off the output, the opposite of the compressor in Set 2 where the heat loss added to the input. The kinetic energy change is under 1 % of the enthalpy drop, which is why it is so often neglected for turbines.

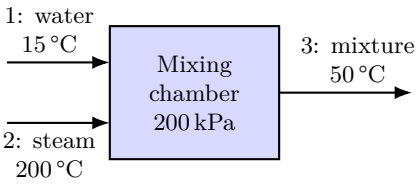
## Question 6 [20 marks]

Cold water at 15 °C enters an insulated mixing chamber at 2.5 kg/s. It is heated by mixing with superheated steam at 200 °C. All three streams are at 200 kPa, and the mixture leaves at 50 °C.

(a) State the phase of each of the three streams and justify it.

(b) Find the mass flow rate of steam that is needed.

(c) The mixture leaves through a pipe with an inside diameter of 5 cm. Find its velocity.



## Solution to Question 6

**Sign convention** The balance is written as energy in equals energy out.

**Given** $P = 200\,\text{kPa}$ for all streams, $T_1 = 15\,^\circ\text{C}$, $\dot{m}_1 = 2.5\,\text{kg/s}$, $T_2 = 200\,^\circ\text{C}$, $T_3 = 50\,^\circ\text{C}$, $D_3 = 0.05\,\text{m}$.

**Find** (the three phases, $\dot{m}_2$, $V_3$)

**Assumptions** Control volume: the mixing chamber, two inlets and one exit. Steady flow. Insulated and no shaft, so $\dot{Q} = \dot{W} = 0$. Kinetic and potential energy changes are negligible.

**Analysis** *(a) Phases.* From Table A-5, the saturation temperature at 200 kPa is 120.21 °C.

- Stream 1: $T_1 = 15\,^\circ\text{C} < T_{\text{sat}}$, compressed liquid.
- Stream 2: $T_2 = 200\,^\circ\text{C} > T_{\text{sat}}$, superheated vapor.
- Stream 3: $T_3 = 50\,^\circ\text{C} < T_{\text{sat}}$, compressed liquid.

*Properties.* A compressed liquid is treated as saturated liquid at the same temperature, so from Table A-4:

$$h_1 \approx h_{\text{f}}(15\,^\circ\text{C}) = 62.982\,\text{kJ/kg}, \qquad h_3 \approx h_{\text{f}}(50\,^\circ\text{C}) = 209.34\,\text{kJ/kg}, \qquad v_3 \approx v_{\text{f}}(50\,^\circ\text{C}) = 0.001012\,\text{m}^3/\text{kg}.$$

From Table A-6 at 0.20 MPa and 200 °C, $h_2 = 2\,870.7\,\text{kJ/kg}$.

*(b) Mass and energy balances.*

$$\text{Mass:} \quad \dot{m}_1 + \dot{m}_2 = \dot{m}_3 \tag{1}$$

$$\text{Energy:} \quad \dot{m}_1 h_1 + \dot{m}_2 h_2 = \dot{m}_3 h_3 \tag{2}$$

Putting (1) into (2) and collecting terms,

$$\dot{m}_1 h_1 + \dot{m}_2 h_2 = (\dot{m}_1 + \dot{m}_2)\, h_3 \quad \Longrightarrow \quad \dot{m}_2 = \dot{m}_1 \frac{h_3 - h_1}{h_2 - h_3}.$$

$$\dot{m}_2 = (2.5\,\text{kg/s})\,\frac{209.34 - 62.982}{2870.7 - 209.34} = (2.5\,\text{kg/s})\,\frac{146.36}{2661.4} = 0.13748\,\text{kg/s}.$$

*(c) Exit velocity.* From (1), $\dot{m}_3 = 2.5 + 0.13748 = 2.6375\,\text{kg/s}$. The pipe area is

$$A_3 = \frac{\pi D_3^2}{4} = \frac{\pi(0.05\,\text{m})^2}{4} = 1.9635 \times 10^{-3}\,\text{m}^2,$$

and from $\dot{m}_3 = V_3 A_3 / v_3$,

$$V_3 = \frac{\dot{m}_3 v_3}{A_3} = \frac{(2.6375\,\text{kg/s})(0.001012\,\text{m}^3/\text{kg})}{1.9635 \times 10^{-3}\,\text{m}^2} = 1.359\,\text{m/s}.$$

(a) 1: compressed liquid, 2: superheated vapor, 3: compressed liquid (b) $\dot{m}_2 = 0.1375\,\text{kg/s}$
(c) $V_3 = 1.359\,\text{m/s}$

**Discussion** Only about 55 g of steam is needed for each kilogram of cold water, because each kilogram of steam gives up 2 661 kJ as it condenses and cools, while each kilogram of water takes only 146.4 kJ to warm by 35 °C. As a check on the compressed-liquid values, $c\,\Delta T = (4.18)(35) = 146.3\,\text{kJ/kg}$, which agrees with $h_3 - h_1$. The exit kinetic energy is $1.359^2/2 = 0.92\,\text{J/kg}$, so neglecting it was safe.